

Sentry.io for Frontend



❖ What Is Sentry?

Sentry is an error-tracking and performance-monitoring tool designed for modern applications. It helps developers identify, diagnose, and resolve issues in real time, providing detailed insights into errors and application performance.

❖ Use of Sentry:

Track Errors Efficiently:

- Detect and log errors in real time, such as unhandled exceptions, crashes, or API failures.
- Automatically capture stack traces to pinpoint where and why errors occur.

Monitor Application Performance:

- Measure key performance metrics like load times, API response times, and user interactions.
- Identify slow pages, routes, or bottlenecks affecting the user experience.

Provide User Context / User Feedback:

- Attach user information to errors to understand how specific users are affected.

Facilitate Team Collaboration:

- Integrate with tools like **Slack**, **Jira**, and **GitHub** to streamline error reporting and resolution workflows.

Support Multiple Platforms:

- Works seamlessly with frontend (React, Angular, Vue) and backend (Node.js, Python, PHP) technologies and many more.

❖ **Benefits of Sentry:**

Proactive Issue Resolution:

- Fix issues before users report them, enhancing user satisfaction.

Increased Developer Productivity:

- Save time debugging with detailed stack traces and breadcrumbs.

Improved Performance:

- Optimize slow-performing pages or features by analysing performance data.

Better User Insights:

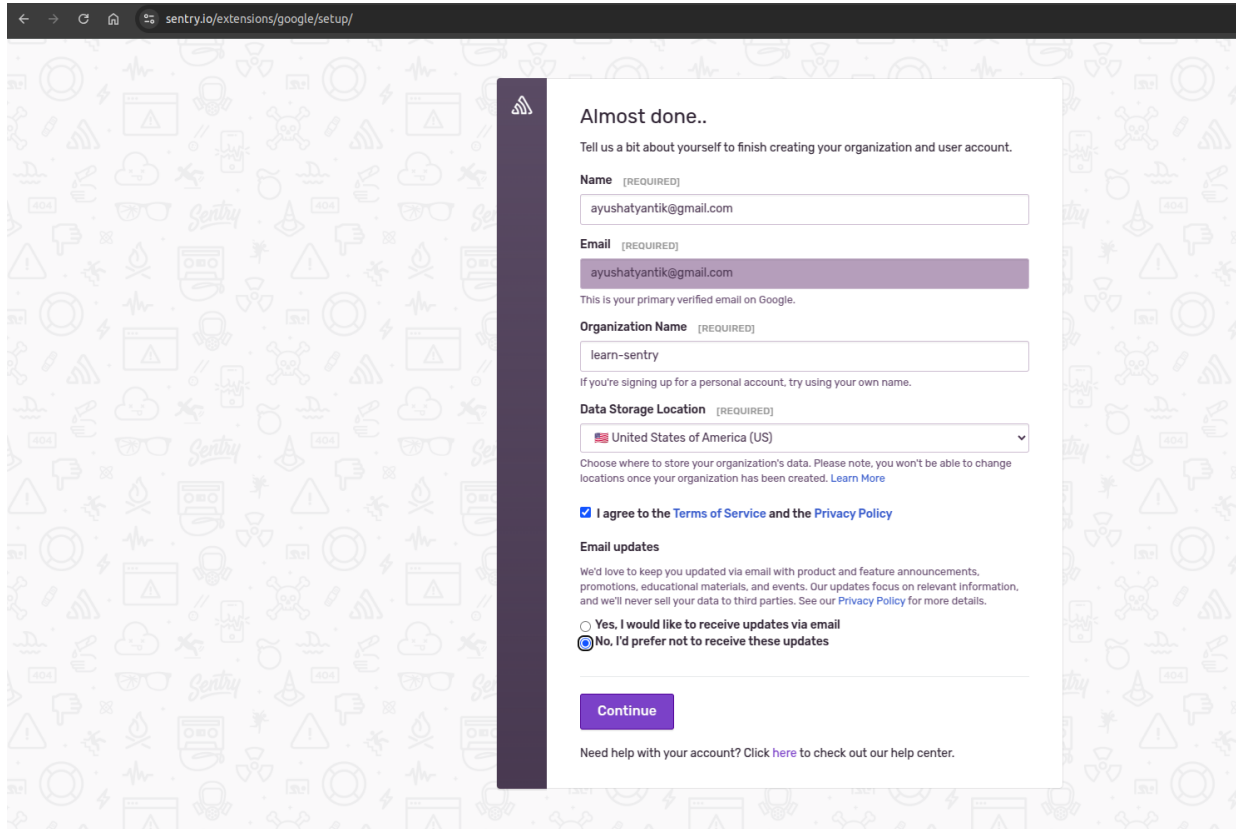
- Understand which issues impact the most users to prioritize fixes effectively.

Scalability:

- Works for projects of all sizes, from small startups to large enterprises.

Sample Project: 1: Frontend Error Monitoring Tutorial

- 1) Clone the repo from below link:
 - <https://github.com/ayushp454/Sentry-React-1>
- 2) Open project and install dependency using below command:
 - npm install
- 3) Login or create your new sentry account. (link: <https://sentry.io/auth/login/>)
I am creating new account... Fill up basic details and click on **Continue**.



The screenshot shows the Sentry account creation form. The background is a light gray pattern of various icons. The form is a white card with a dark purple header bar on the left containing the Sentry logo. The form title is "Almost done.." and the subtitle is "Tell us a bit about yourself to finish creating your organization and user account." The form fields are: "Name" (required) with the value "ayushatyantik@gmail.com", "Email" (required) with the value "ayushatyantik@gmail.com", "Organization Name" (required) with the value "learn-sentry", and "Data Storage Location" (required) with a dropdown menu showing "United States of America (US)". Below the "Data Storage Location" field, there is a note: "Choose where to store your organization's data. Please note, you won't be able to change locations once your organization has been created. [Learn More](#)". There is a checkbox "I agree to the [Terms of Service](#) and the [Privacy Policy](#)" which is checked. Below this, there is a section for "Email updates" with the text: "We'd love to keep you updated via email with product and feature announcements, promotions, educational materials, and events. Our updates focus on relevant information, and we'll never sell your data to third parties. See our [Privacy Policy](#) for more details." There are two radio buttons: "Yes, I would like to receive updates via email" (unchecked) and "No, I'd prefer not to receive these updates" (checked). At the bottom of the form is a purple "Continue" button. Below the button, there is a link: "Need help with your account? Click [here](#) to check out our help center."

Almost done..

Tell us a bit about yourself to finish creating your organization and user account.

Name [REQUIRED]
ayushatyantik@gmail.com

Email [REQUIRED]
ayushatyantik@gmail.com

This is your primary verified email on Google.

Organization Name [REQUIRED]
learn-sentry

If you're signing up for a personal account, try using your own name.

Data Storage Location [REQUIRED]
United States of America (US)

Choose where to store your organization's data. Please note, you won't be able to change locations once your organization has been created. [Learn More](#)

☒ I agree to the [Terms of Service](#) and the [Privacy Policy](#)

Email updates

We'd love to keep you updated via email with product and feature announcements, promotions, educational materials, and events. Our updates focus on relevant information, and we'll never sell your data to third parties. See our [Privacy Policy](#) for more details.

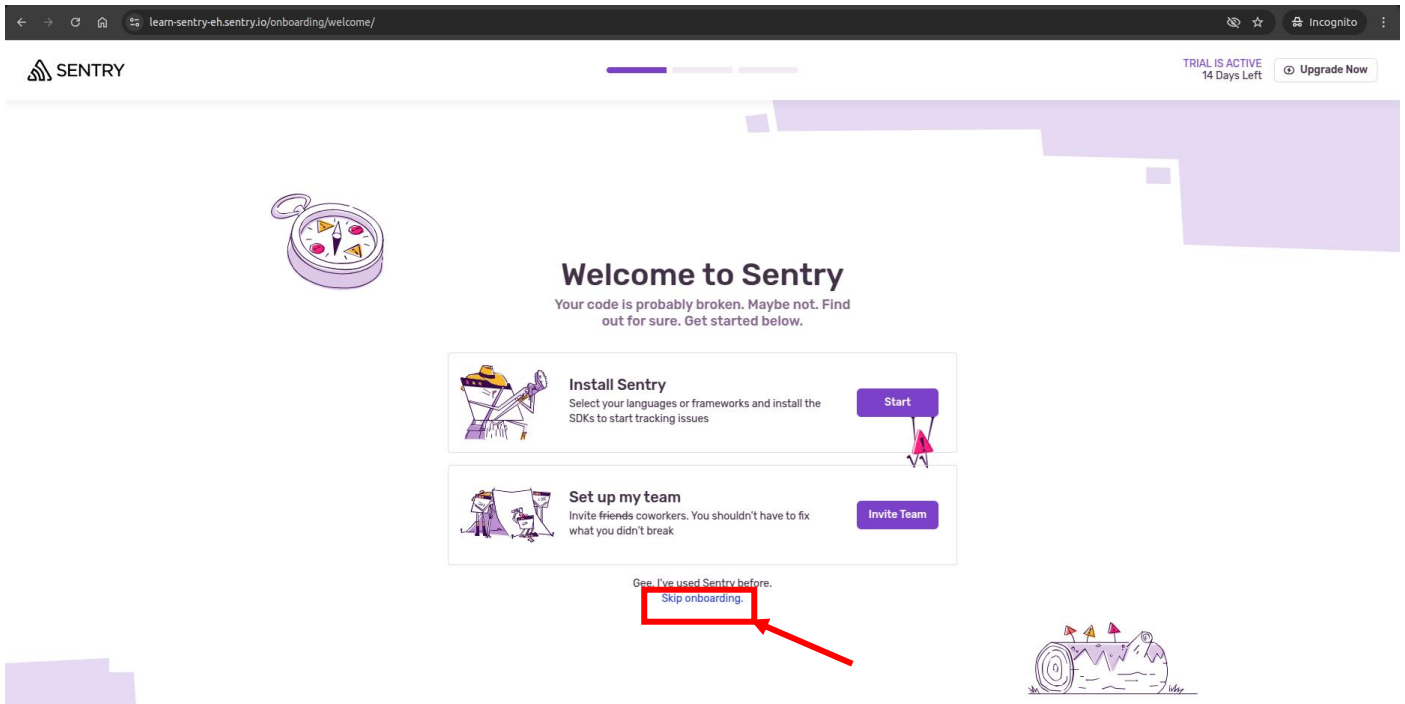
☐ Yes, I would like to receive updates via email

☒ No, I'd prefer not to receive these updates

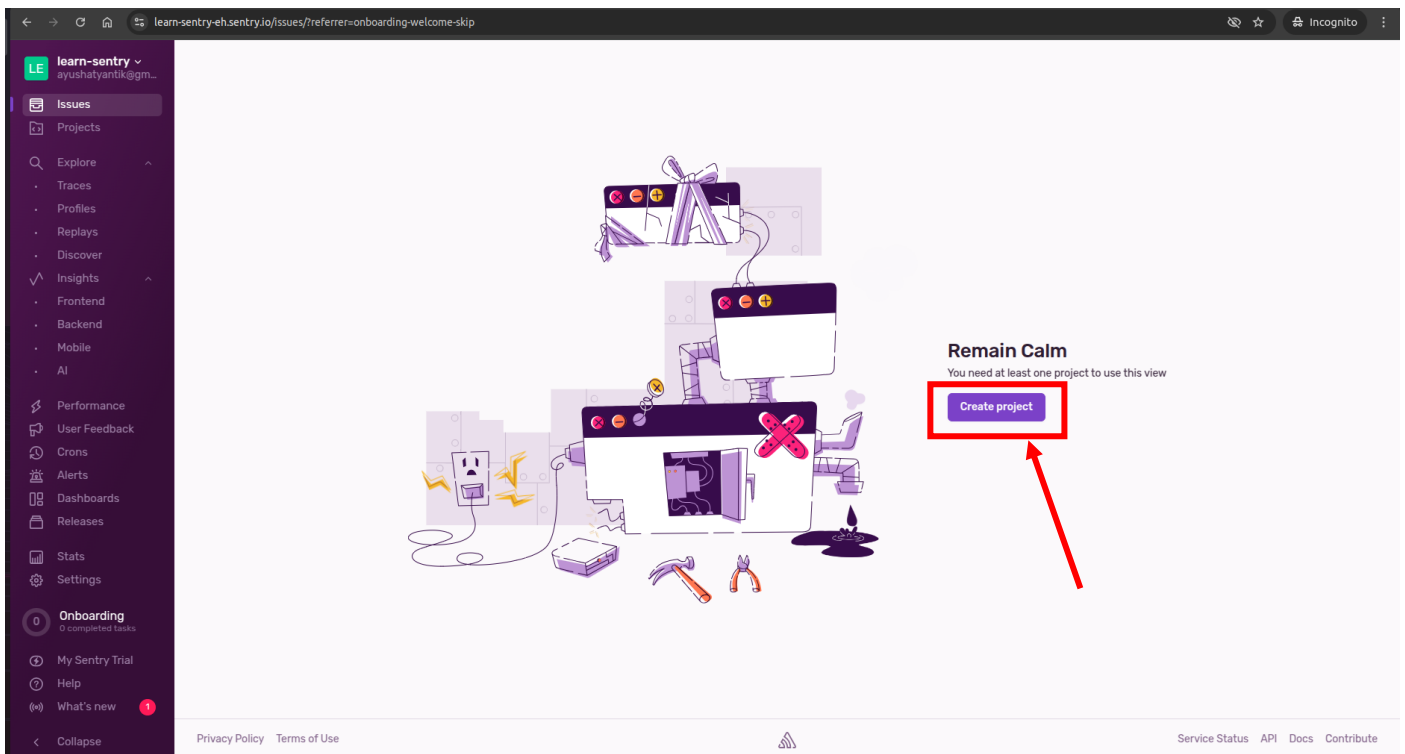
[Continue](#)

Need help with your account? Click [here](#) to check out our help center.

4) Click on **Skip on boarding.**



5) Now, you can show below dashboard & create your first Project.



6) Project basic configuration.

- **Choose your platform:** React JS
- **Set your alert frequency:** Alert me on high priority issues (*default*)
- **Name your project and assign it a team:**
 - **Project name:** sentry-frontend-1
 - **Team:** #learn-sentry (*default*)

Click on **Create Project**.

learn-sentry-eh.sentry.io/projects/new/

Create a new project in 3 steps

Set up a separate project for each part of your application (for example, your API server and frontend client), to quickly pinpoint which part of your application errors are coming from. [Read the docs.](#)

- Choose your platform**
Popular Browser Server Mobile Desktop Serverless All
- Set your alert frequency**
 - ☒ Alert me on high priority issues **2**
 - ☐ When there are more than occurrences of a unique error in
 - ☐ I'll create my own alerts later
 - ☒ Notify via email
 -
- Name your project and assign it a team** **3 & 4**
 - Project name
 - Team
 - 5**

Privacy Policy Terms of Use Service Status API Docs Contribute

7) Now, you can show set up screen.

Configure React SDK

In this quick guide you'll use `npm` or `yarn` to set up:

- ☒ Error Monitoring
- ☒ Tracing
- ☒ Session Replay

Install

Add the Sentry SDK as a dependency using `npm` or `yarn`:

```
npm yarn
npm install --save @sentry/react
```

Configure SDK

Initialize Sentry as early as possible in your application's lifecycle.

```
JavaScript
import * as Sentry from "@sentry/react";

Sentry.init({
  dsn: "https://315440216aa9e97f93f7b922fec6fb0e4508657510514688@ingest.us.sentry.io/4508657572184064",
  integrations: [
    Sentry.browserTracingIntegration(),
    Sentry.replayIntegration(),
  ],
  // Tracing
  tracesSampleRate: 1.0, // Capture 100% of the transactions
  // Set 'tracePropagationTargets' to control for which URLs distributed tracing should be enabled
  tracePropagationTargets: ["localhost", /^https:\/\/yourserver\.io\/api/],
  // Session Replay
  replaysSessionSampleRate: 0.1, // This sets the sample rate at 10%. You may want to change it to 100% while in development and then sample at a lower rate in production.
  replaysOnErrorSampleRate: 1.0, // If you're not already sampling the entire session, change the sample rate to 100% when sampling sessions where errors occur.
});

const container = document.getElementById("app");
const root = createRoot(container);
root.render(<App />);
```

Upload Source Maps

Automatically upload your source maps to enable readable stack traces for Errors. If you prefer to manually set up source maps, please follow [this guide](#).

```
npm @sentry/wizard@latest --source-maps --saas
```

Verify

This command will show an intentional error and can be used as a test to make sure that everything is working as expected.

8) Copy below command and open your project terminal and paste it.

- `npm install --save @sentry/react`

```
ayush@ayush-Vostro-3520:~/Desktop/Practice Projects/Sentry-Practice/Sentry-1/Sentry-project-1(develop)$ npm install --save @sentry/react
added 9 packages, and audited 268 packages in 3s

107 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
```

9) Copy **configure SDK** code and paste that code in project `src/main.tsx` file.

Configure SDK

Initialize Sentry as early as possible in your application's lifecycle.

```
JavaScript

import * as Sentry from "@sentry/react";

Sentry.init({
  dsn: "https://315440216aa9e97f93f7b922fecefb6b04508657510514688@ingest.us.sentry.io/4508657572184064",
  integrations: [
    Sentry.browserTracingIntegration(),
    Sentry.replayIntegration(),
  ],
  // Tracing
  tracesSampleRate: 1.0, // Capture 100% of the transactions
  // Set 'tracePropagationTargets' to control for which URLs distributed tracing should be enabled
  tracePropagationTargets: ["localhost", /^https:\/\/yourserver\.io\/api$/],
  // Session Replay
  replaysSessionSampleRate: 0.1, // This sets the sample rate at 10%. You may want to change it to 100% while in development and then sample at a lower rate in production.
  replaysOnErrorSampleRate: 1.0, // If you're not already sampling the entire session, change the sample rate to 100% when sampling sessions where errors occur.
});

const container = document.getElementById("app");
const root = createRoot(container);
root.render(<App />);
```

- Paste this code in `src/main.tsx` file. ([see here code changes](#))

10) In project goto `src/App.jsx` file and un comment the line number 6.

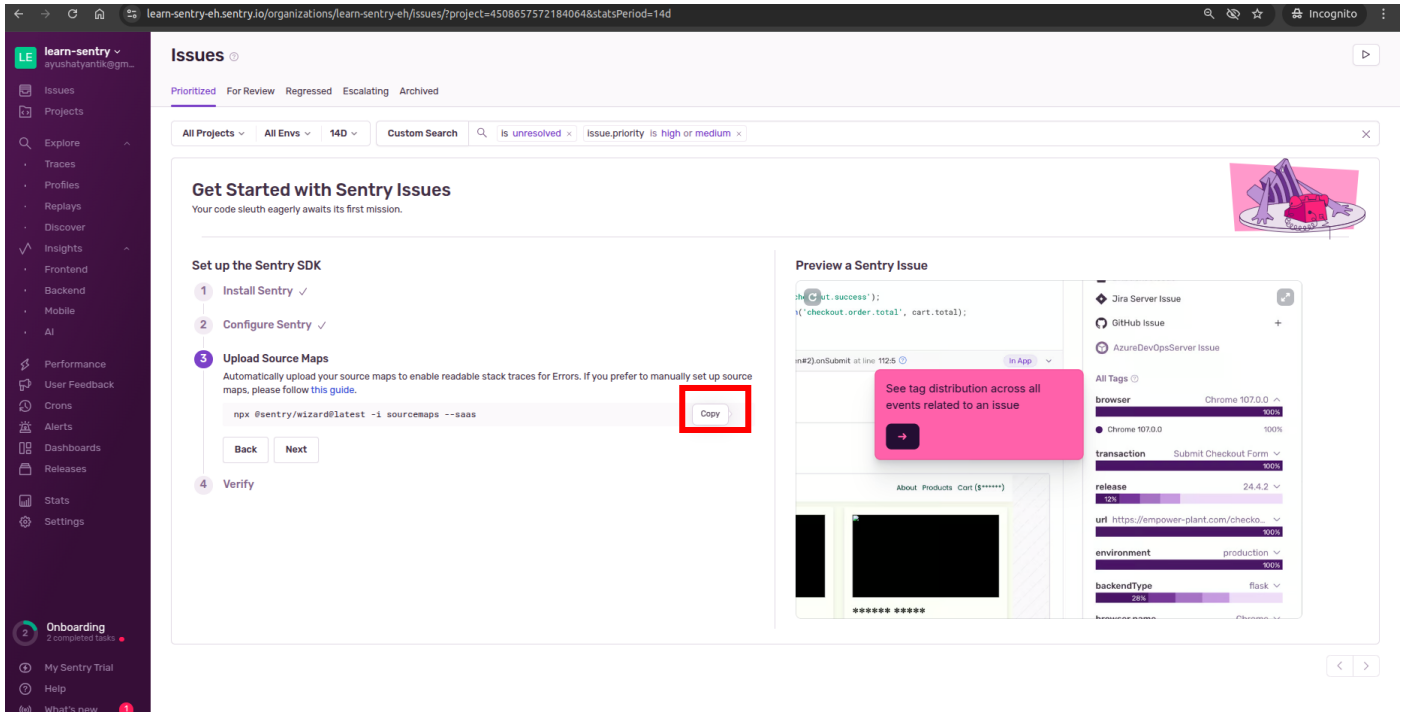
- ([See here code changes](#))

11) In Sentry click on **Take me to Issues** button. Or go to Issues pages.

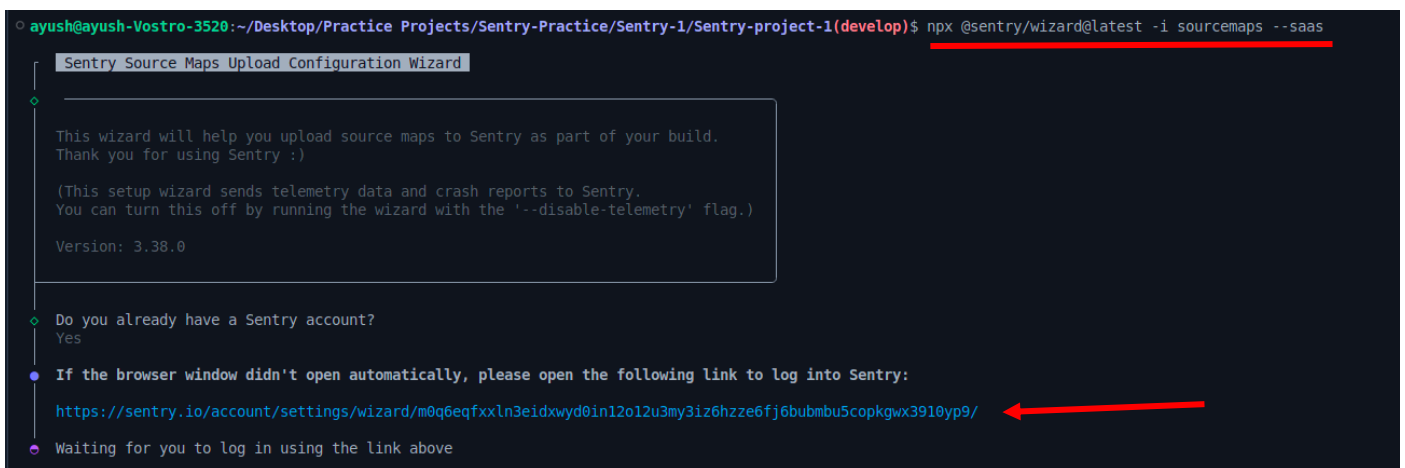
The screenshot shows the Sentry web interface. On the left sidebar, the 'Issues' button is highlighted with a red box. The main content area displays the 'Configure SDK' page with a code block for JavaScript. A red arrow points to the 'Copy' button in the top right corner of the code block. Below the code block, there are sections for 'Upload Source Maps', 'Verify', and 'Additional Information'. At the bottom of the main content area, the 'Take me to Issues' button is highlighted with a red box.

12) Upload your source map.

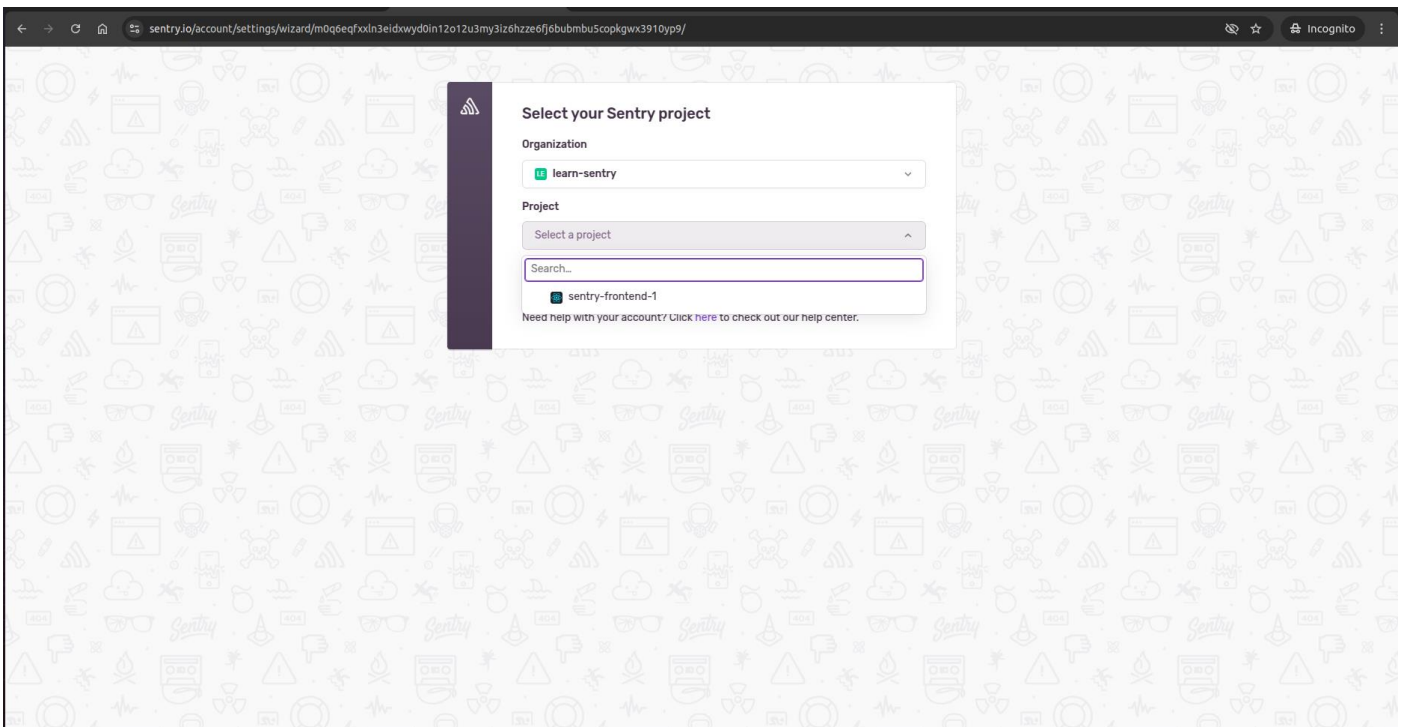
- A source map is a valuable file that helps identify the exact line of code where an error occurred. It also provides a detailed stack trace, making it easier to debug and resolve issues efficiently.
- For uploading source map, you can also follow this doc: [Manually setup source map](#)
- Copy below terminal command.
→ `npx @sentry/wizard@latest -i sourcemaps --saas`



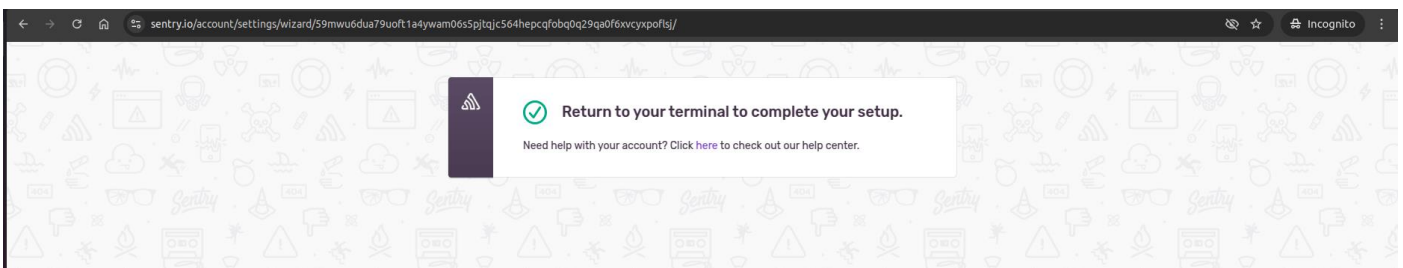
- Paste this command in your project level root folder terminal.



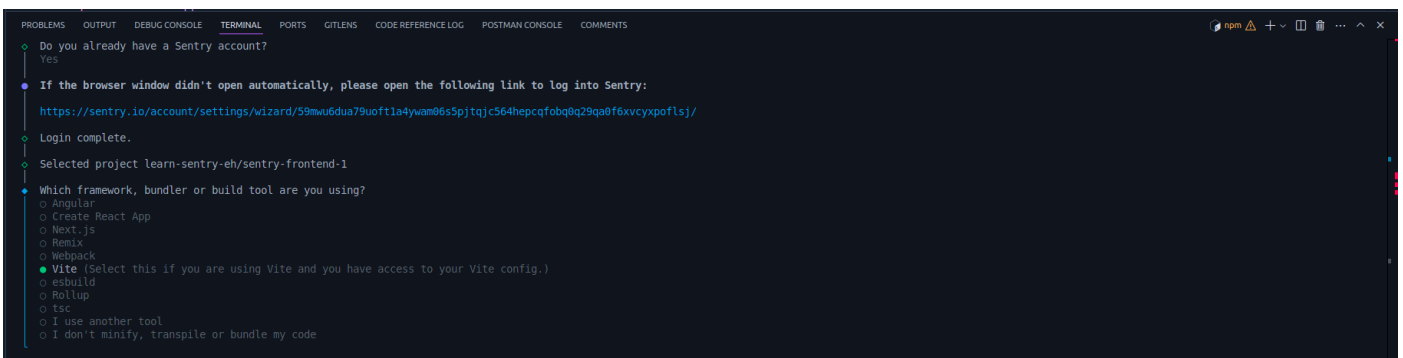
- Open link which shows in terminal.



- Select **sentry-frontend-1** and continue.
- You will see below screen. & go back to your terminal.



- Select Framework/bundler or build tool: **Vite**



- Are you using CI/CD tool to build and deploy your application ? --> **No**


```
ayush@ayush-Vostro-3520:~/Desktop/Practice Projects/Sentry-Practice/Sentry-1/Sentry-project-1(develop)$ npx @sentry/wizard@latest -i sourcemaps --saas
```

Sentry Source Maps Upload Configuration Wizard

This wizard will help you upload source maps to Sentry as part of your build.
Thank you for using Sentry :)

(This setup wizard sends telemetry data and crash reports to Sentry.
You can turn this off by running the wizard with the '--disable-telemetry' flag.)

Version: 3.38.0

- ◊ Do you already have a Sentry account?
Yes
- If the browser window didn't open automatically, please open the following link to log into Sentry:
<https://sentry.io/account/settings/wizard/59mwu6dua79uoft1a4ywam06s5pjtqjc564hepcqfobq0q29qa0f6xvcyxpoflsj/>
- ◊ Login complete.
- ◊ Selected project learn-sentry-eh/sentry-frontend-1
- ◊ Which framework, bundler or build tool are you using?
Vite
- ◊ Installed @sentry/vite-plugin with NPM.
- Added the Sentry Vite plugin to vite.config.js and enabled source maps
- We recommend checking the modified file after the wizard finished to ensure it works with your build setup.
- Created .env.sentry-build-plugin with auth token for you to test source map uploading locally.
- Added .env.sentry-build-plugin to .gitignore.
- Are you using a CI/CD tool to build and deploy your application?
 - Yes
 - No (I build and deploy my application manually)

- After this you can show below message from terminal.

```
◊ Are you using a CI/CD tool to build and deploy your application?  
No  
● No Problem! Just make sure that the Sentry auth token from .env.sentry-build-plugin is available whenever you build and deploy your app.  
That's it - everything is set up!  
Test and validate your setup locally with the following Steps:  
1. Build your application in production mode.  
→ For example, run npm run build.  
→ You should see source map upload logs in your console.  
2. Run your application and throw a test error.  
→ The error should appear in Sentry:  
→ https://learn-sentry-eh.sentry.io/issues/?project=4508657572184064  
3. Open the error in Sentry and verify that it's source-mapped.  
→ The stack trace should show your original source code.  
  
If you encounter any issues, please refer to the Troubleshooting Guide:  
https://docs.sentry.io/platforms/javascript/sourcemaps/troubleshooting_js  
  
If the guide doesn't help or you encounter a bug, please let us know:  
https://github.com/getsentry/sentry-javascript/issues
```

- This process also add some changes in code. ([see changes here](#))
 - File: .gitignore
 - # Sentry Config File
 - .env.sentry-build-plugin
 - File: package.json

```
"dependencies": {  
  ...  
  "@sentry/vite-plugin": "^3.0.0",  
  ...  
},
```

- File: vite.config.js

13) Build your react application using below command.

- npm run build

```
ayush@ayush-Vostro-3520:~/Desktop/Practice Projects/Sentry-Practice/Sentry-1/Sentry-project-1(develop)$ npm run build
> sentry-project-1@0.0.0 build
> vite build

[sentry-vite-plugin] Info: Using environment variables configured in ".env.sentry-build-plugin".
vite v6.0.7 building for production...
[sentry-vite-plugin] Info: Sending telemetry data on issues and performance to Sentry. To disable telemetry, set `options.telemetry` to `false`.
✓ 251 modules transformed.
dist/index.html      0.46 kB | gzip: 0.29 kB
dist/assets/index-n_ryQ3BS.css 1.39 kB | gzip: 0.71 kB
dist/assets/index-CowbaGKf.js 383.83 kB | gzip: 125.52 kB | map: 1,582.19 kB
> Found 2 files
> Analyzing 2 sources
> Analyzing completed in 0.014s
> Adding source map references
> Bundling completed in 0.185s
> Bundled 2 files for upload
> Bundle ID: 0e84e99c-cd24-5402-9162-418e8da34a52
> Optimizing completed in 0.007s
> Uploading completed in 0.895s
> Uploaded files to Sentry
> Processing completed in 0.387s
> File upload complete (processing pending on server)
> Organization: learn-sentry-eh
> Project: sentry-frontend-1
> Release: 66f8d630d6e319557b7da64d94d47837c8c30214
> Dist: None
> Upload type: artifact bundle

Source Map Upload Report
Scripts
~/ada54132-754b-4ba6-a7e3-877178b933d1-0.js (sourcemap at index-CowbaGKf.js.map, debug id ada54132-754b-4ba6-a7e3-877178b933d1)
Source Maps
~/ada54132-754b-4ba6-a7e3-877178b933d1-0.js.map (debug id ada54132-754b-4ba6-a7e3-877178b933d1)
[sentry-vite-plugin] Info: Successfully uploaded source maps to Sentry
✓ built in 8.39s
```

You can see in the terminal it successfully adds sentry configuration with a source map.

14) Run your project by following below command.

- npm run preview

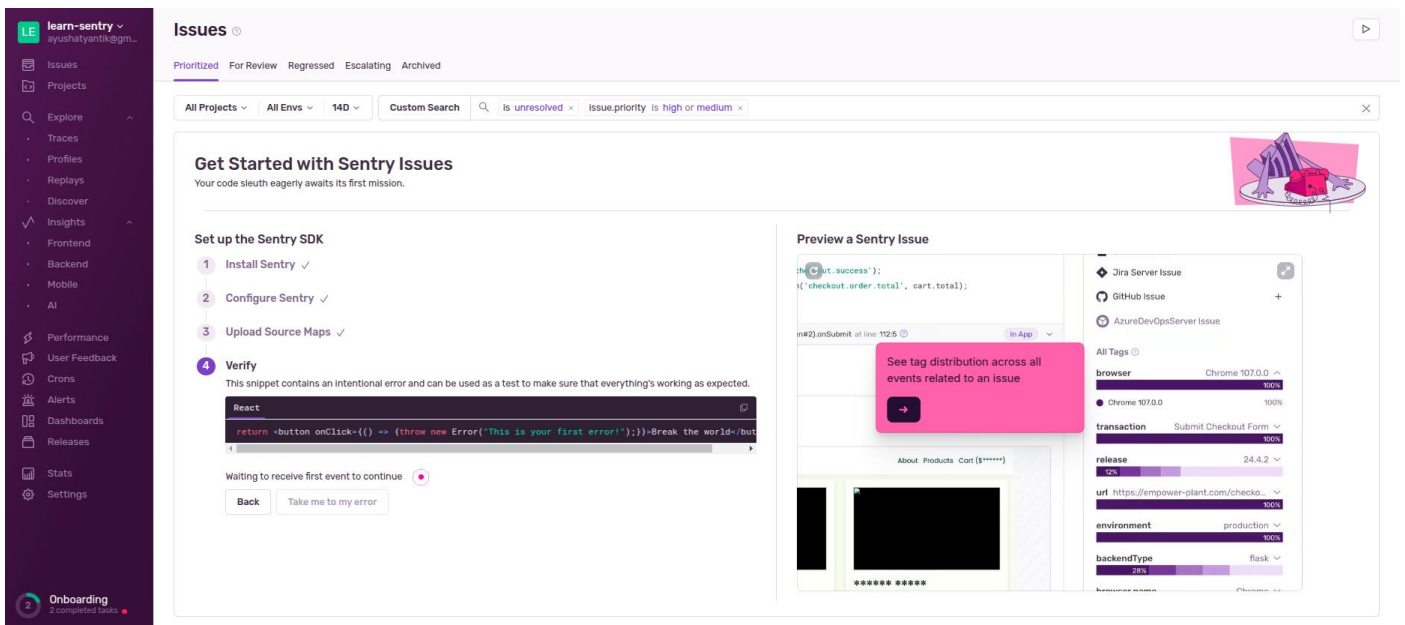
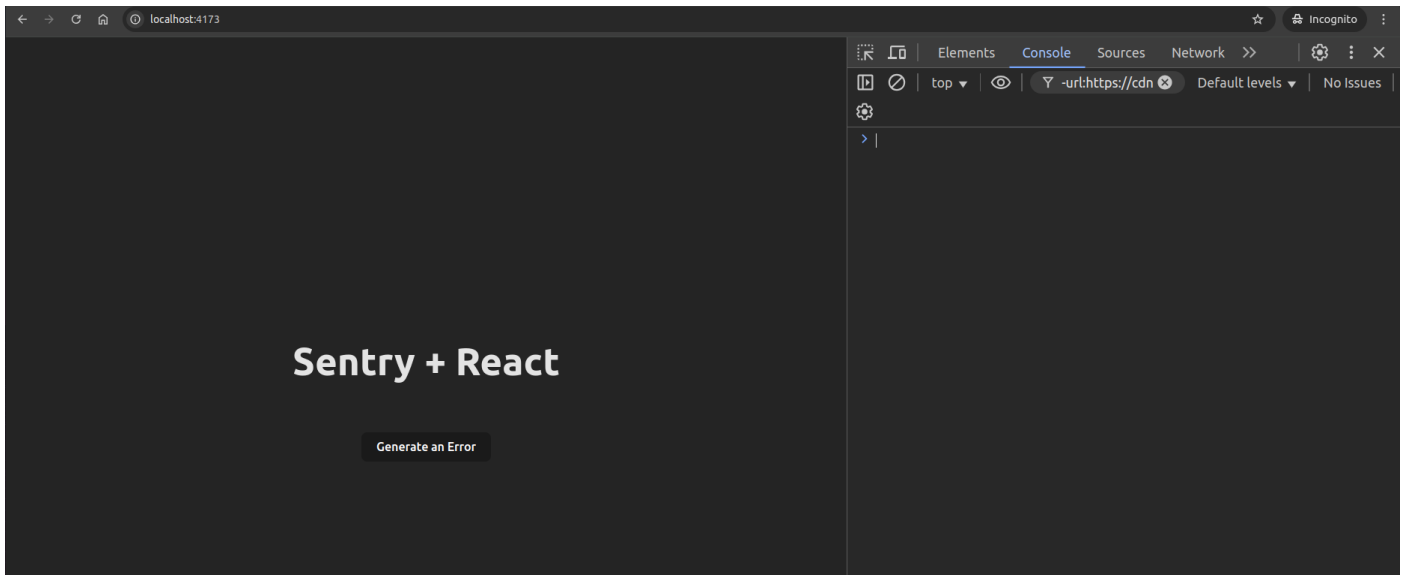
```
ayush@ayush-Vostro-3520:~/Desktop/Practice Projects/Sentry-Practice/Sentry-1/Sentry-project-1(develop)$ npm run preview
> sentry-project-1@0.0.0 preview
> vite preview

✗ vite v6.0.7 is not yet supported in the Community edition of Console Ninja.
We are working hard on it for you https://tinyurl.com/3h9mtwra.

Estimated release dates:
- Community users: around 2nd March, 2025 (subject to team availability)
- PRO users:      priority access is available now

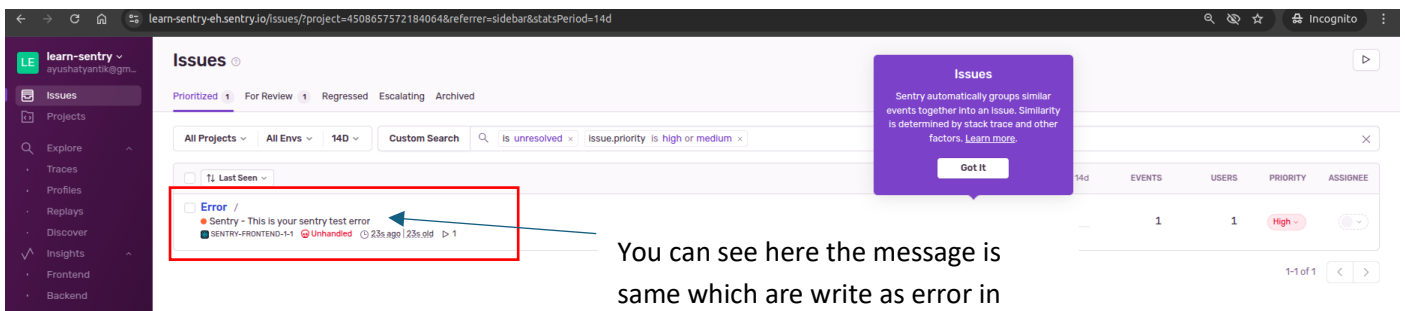
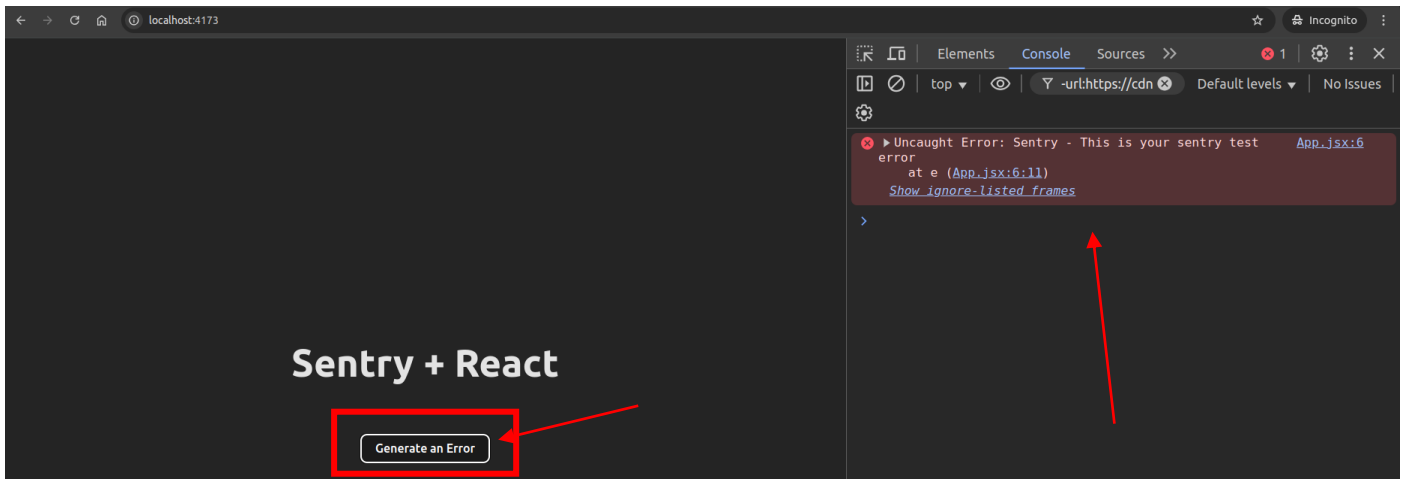
[sentry-vite-plugin] Info: Using environment variables configured in ".env.sentry-build-plugin".
→ Local: http://localhost:4173/
→ Network: use --host to expose
→ press h + enter to show help
```

15) Open this link in your browser: <http://localhost:4173/>

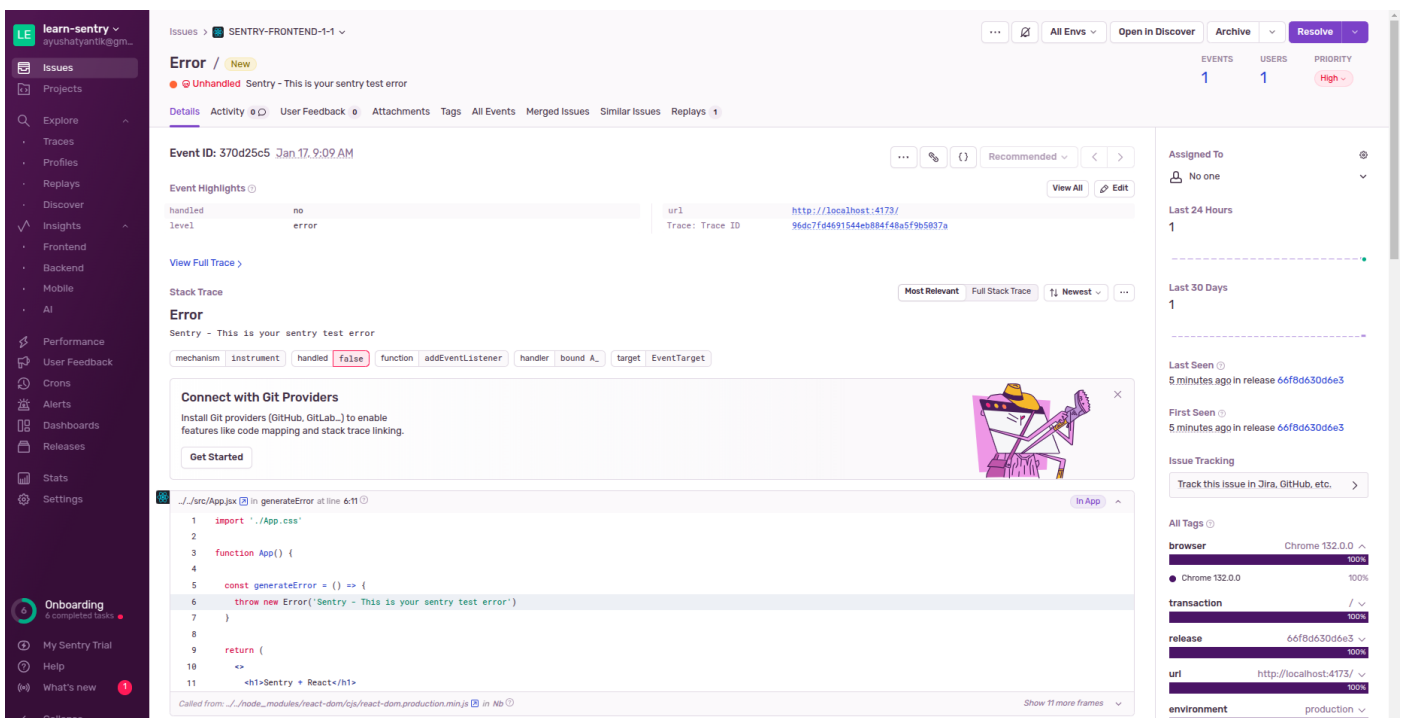


16) Click on **Generate an Error** button.

It will generate an error and directly log into sentry system. In the **Issue page**.

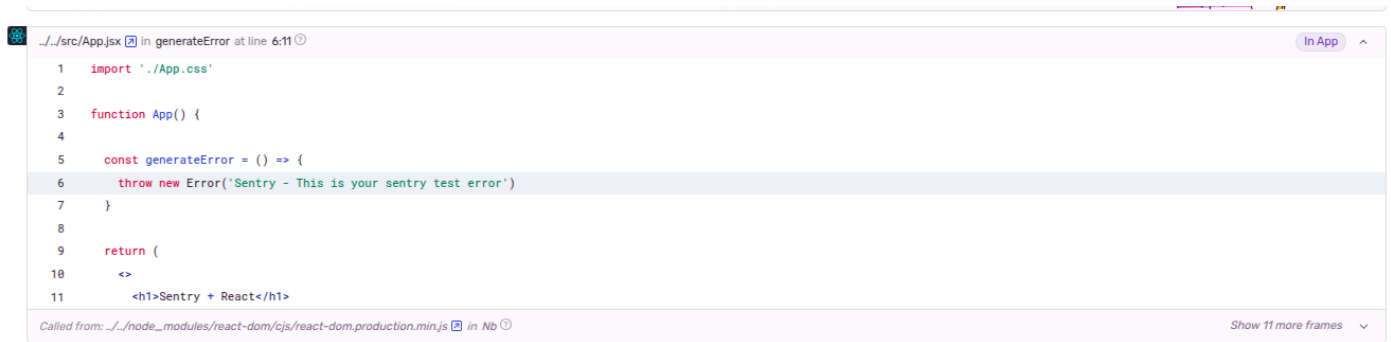


17) Open that error.



- In that screen we can see all the detail about that error

- i) Below sections show us in which line we get an error. & it exact same line of our code which we write. (It's a stack trace section)



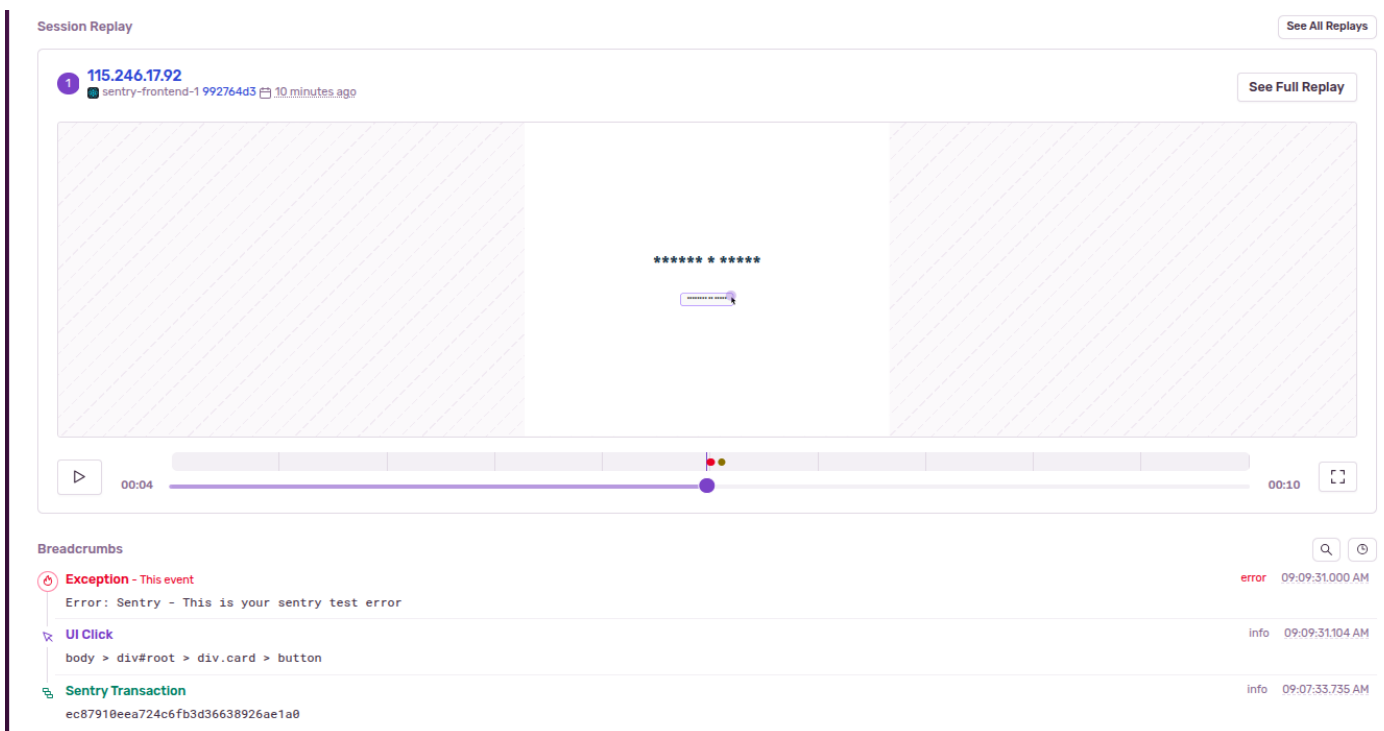
```
1 import './App.css'
2
3 function App() {
4
5   const generateError = () => {
6     throw new Error('Sentry - This is your sentry test error')
7   }
8
9   return (
10     <>
11     <h1>Sentry + React</h1>
12   )
13 }
```

Called from: .../node_modules/react-dom/cjs/react-dom.production.min.js in Nb

Show 11 more frames

- ii) In below section we show **Reply**

It show us how error generated on frontend side. You can play that video.



Session Replay

115.246.17.92 sentry-frontend-1 992764d3 10 minutes ago

00:04 00:10

Breadcrumbs

- Exception - This event
Error: Sentry - This is your sentry test error
error 09:09:31.000 AM
- UI Click
body > div#root > div.card > button
info 09:09:31.104 AM
- Sentry Transaction
ec87910eea724c6fb3d36638926ae1a0
info 09:07:33.735 AM

- iii) Below details represent how many (**user**) users face this error and (**events**) how many time they generate this same error.

EVENTS	USERS	PRIORITY
2	1	High

- iv) Below screen shot show us we can assign this issue to our team mates, time line of issue, last seen of this issue face, Issue tracking: in which we can track issue by directly creating the jira ticket, generate issue in GitHub, etc.

Assigned To 

 No one 

Last 24 Hours

1



Last 30 Days

1



Last Seen 

13 minutes ago in release [66f8d630d6e3](#)

First Seen 

13 minutes ago in release [66f8d630d6e3](#)

Issue Tracking

Track this issue in Jira, GitHub, etc. 

Usefull links:

[Frontend Error Monitoring tutorial](#)

[Sourc Map - enable readable stack traces in your Sentry errors.](#)

[Sentry React feature](#)

[React router - sentry \(version wise\)](#)

[Basic configuration Options.](#)